

Bappy Chandra Debnath

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EDUCATION

Ahsanullah University of Science and Technology

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering (CGPA: 3.282/4.00)

June 2026

Relevant Coursework: AI Engineering, LLM Fine-tuning, AI Agents, Software Engineering Principles, Full-stack Development

PROJECTS & EXPERIENCE

OmniCart AI — Multi-Agent Swarm for E-commerce

Personal Project

- Architected a LangGraph-powered multi-agent system managing autonomous customer query routing, live SQLite inventory tracking, and vector-based policy retrieval (ChromaDB RAG).
- Integrated Llama-3.2-Vision for instant product image identification and structured context extraction.
- Crafted an interactive Streamlit Admin Dashboard featuring real-time chat history sync and a "Human-in-the-Loop" (HITL) take-over trigger mechanism.

AgroPulse-BD — Deep Learning & Remote Sensing

Project Link | Lead Developer

- Engineered a predictive decision support system utilizing Bi-LSTM and 1D-CNN architectures for high-precision flood forecasting and crop health monitoring in the Sunamganj region.
- Synthesized 15 years of multi-modal data, integrating Sentinel-1 SAR and Sentinel-2 NDVI datasets via daily scheduled GitHub Actions pipelines.
- Optimized deep learning models using Focal Loss and implemented model explainability through SHAP; deployed Google Gemini via API for localized Bengali warnings.

AI-Powered Gmail Agent — RAG System

Project Lead / Developer

- Spearheaded a RAG-driven AI assistant using Groq (Llama 3) and Sentence Transformers to generate professional replies from a Supabase vector database.
- Implemented a robust Human-in-the-Loop workflow with a 1-5 star operator rating feedback loop to refine future prompts dynamically.

Smart Office IoT Monitor — Full-Stack IoT & AI SaaS

Project Link | Independent Project

- Designed a full-stack smart building system integrating an **Arduino Uno** simulation with a responsive **Vanilla JavaScript** dashboard for real-time, bidirectional device control.
- Engineered a **FastAPI** backend to manage device states, compute live power consumption, and execute automated after-hours alerts.
- Integrated **Gemini 2.5 Flash** via **Discord.py** to parse system telemetry and generate natural language status reports.

AI Call Center Analytics — Full-Stack AI SaaS

Live Demo | Independent Project

- Engineered an automated processing pipeline using **FastAPI** and **Gemini 1.5 Flash** to analyze live support calls, extracting agent performance, sentiment, and business insights into strict JSON data.
- Implemented a real-time data architecture with **Supabase** alongside a smart queue system to eliminate data duplication and handle asynchronous cloud workloads seamlessly.
- Designed a lightweight frontend using **Vanilla JavaScript** featuring a live UI terminal log for real-time tracking and a 1-click PDF generation engine for instant reporting.

LEADERSHIP & ACTIVITIES

Ahsanullah University of Science & Technology

Mar 2022 – Present

- Led cross-functional development teams for major projects, including AgroPulse-BD, coordinating model implementation and deployment.
- Directed full-stack development efforts, backend architecture planning, and collaborated with faculty mentors on multi-modal data fusion research.

SKILLS & INTERESTS

Programming & Tools: Python, C++, C, Django, PHP, SQL, n8n, FastAPI, PySpark, Git Hygiene, GitHub Actions

AI/ML & Architecture: LangGraph, LangChain, Deep Learning (Bi-LSTM, 1D-CNN), RAG, ChromaDB, Supabase, LLM Fine-tuning

Problem Solving: Solved 100+ problems across LeetCode, NeetCode, Codeforces, & AtCoder

Frontend & UI/UX: Streamlit, Modern HTML, CSS, UI/UX Design

Interests: Agentic AI, Geospatial Analytics, Human-AI Collaboration